The symbols refer to those given above and in Fig. A.4. Mass flow rates and conditions inside each battery can in principle be obtained recursively from the relations

$$\xi_{n+1} = \xi_n(1 + P_n) - \eta_n \quad \text{and} \quad \eta_{n+1} = \xi_n P_n \tag{A.18}$$

However, numerical problems tend to make the use of these equations difficult for batteries with many stages. Special cases occur when P is either constant or constantly equal 1.

The flow rates out from any stage can usually be calculated with sufficient accuracy from the apparent molar volumes, Φ_i, of the i compounds according to

$$V_{aq} = V_{0,aq} \sum \left(1 + \Phi_{i,aq} X_i\right) \quad \text{and} \quad V_{org} = V_{0,org} \sum \left(1 + \Phi_{i,org}\, Y_i\right) \tag{A.19}$$

where $V_{0,aq}$ and $V_{0,org}$ are the flow rates of the pure unloaded phases in each stage. In many cases one of the central feeds is missing (usually the organic feed) and the corresponding mass and volume flow is zero. The wash stream (index W) is then the real feed for that phase.

When the flow rates vary considerably in the batteries due to loading effects, the whole calculational procedure becomes recursive and a computer code based on these equations may be needed. However, in most cases it is only necessary to treat one, or perhaps two, of the extracted components recursively as it dominates volume changes and defines the P-values for all minor components, e.g. in the high active first stage in the Purex process nitric acid and uranium define the flow rates and extraction factors for all other elements.

In industrial processes it is not only important to extract the desired compound with high yield and purity, but also to be able to strip it again with good efficiency using a minimum amount of chemicals. Contrary to single stage batch extraction, the possibility to vary the P-value by changes in the flow rate ratio (remember that $P = D \times V_{org}/V_{aq}$) makes the use of conditions with D-values for the product not too far from 1 is desirable. In such a case the product can be stripped by a decrease in the flow rate ratio (in more dilute form than in the feed) and a moderate decrease in the D-value without the need to introduce new chemicals for stripping. A typical example is the uranium purification part of the Purex process. Here uranium (and nitric acid) is extracted at high concentration in the presence of 1−6 M HNO_3 and stripped at a low flow rate ratio by more dilute nitric acid. If needed, the separation can be repeated after reconcentration by evaporation.

A.4. SOLVENT EXTRACTION EQUIPMENT

The less conventional part − from a chemical engineering viewpoint − of a reprocessing plant is the solvent extraction equipment, even though the technique is becoming

increasingly common in the chemical industry. The principle of all such equipment is illustrated by Fig. A.1(a); it contains a mixing section for efficient transfer of materials between the phases and a settling section for efficient phase separation. The inputs and outputs provided for connecting stages in the counter current extraction scheme (Fig. A.2). *Mixer-settlers* (Fig. A.1(a)) provide good mixing and reasonably good phase separation performance but rather large hold-ups.

For reprocessing the equipment must be highly reliable, have high stage efficiency, short contact times, small liquid inventory (hold-up), be easy to decontaminate and to service, and not least be safe against criticality. High reliability usually means simple design and few (if any) moving parts. The *packed columns* meet this requirement. They are simply long columns (often 10–20 m with a diameter of 0.3–3 m) filled with small pieces of material obstructing a straight flow through the column, which is by gravity (Fig. A.1(b)). However, high stage efficiency requires mechanical agitation of the two phases and clean phase separation, which cannot be met by packed columns.

In *pulsed columns* (Fig. A.1(c)) the mechanical agitation provides good mixing but poor phase separation. Each plate is perforated (a sieve). The organic and aqueous phases separate between the plates ('settling chambers'). In the down movement (<¼ of the interplate distance) the aqueous phase is forced through the sieves, forming droplets, which by gravity fall through the lighter organic phase and coalesce when reaching the interface boundary. In the upward stroke, organic droplets form and rise through the aqueous phase until they meet the organic phase boundary.

The phase separation in a mixer-settler battery or in a column is usually not better than ~99%, i.e. each outgoing phase contains some percent entrained droplets of the other phase. This separation efficiency can be improved to almost 100% by using *centrifugal extractors* (Fig. A.1(d)). Centrifugal extractors affect good mixing, good phase separation, and have very small hold-ups. The organic-aqueous phase contact time in centrifugal extractors can be made much shorter than in mixer-settlers or columns. The small hold-up volume and short residence time cooperate to reduce radiation decomposition.

Packed columns were used in the first Windscale plant (Sellafield, UK). Pulsed columns were used at Hanford (USA), in the old Eurochemic plant at Mol (Belgium), and are currently in use in the newer La Hague and THORP plants. Mixer-settlers have been used at Savannah River (USA), in the Magnox plant at Sellafield (UK), and at La Hague (France). Centrifugal extractors have been installed at Savannah River and at La Hague.

A.5. EXERCISES

A.1. In a solvent system the distribution ratio, D_U, is 2 for uranium and D_{Cs} is 0.003 for caesium. If 99.5% U is to be extracted in a repeated batch fashion (Eqn (A.5)), (a) how much Cs is coextracted? If instead a counter current process is used with